Changing a Networked Mind: Paradigm Dynamics as Structure Learning over a Hidden World

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Abstract. How does a scientific community change its mind—not about a fact, but about the *frame* its facts live in? We model a paradigm as a directed network of commitments and a community as bounded active-inference agents who learn that network's structure while valuing some conclusions over others and pooling beliefs with peers. Two classical theorems make this community a null regime: Gaussian beliefs under precision-weighted pooling can neither reject a conflicting source nor sustain a school. Result 1 measures the consequence, the *homogeneity effect*: communities that would learn two crisply different wirings apart collapse, under any contact, into one compromise net no member holds. Result 2 measures the repair, in two stages: infer each source's *reliability* from surprise and distinctness returns as a timescale; *represent the rival*—candidate wirings, a posterior over them, hypotheses communicated rather than averaged—and it returns as a wall, two internally certain camps at a genuine fixed point. Gates buy time; represented rivals buy walls. Averaging refines a frame; it cannot grow a new one.

Keywords: Active inference · Structure learning · Belief sharing · Paradigm change · Social epistemology

1 Introduction: changing a frame, not a credence

If we want science to keep going well, we need to know how it *works*: what kind of agent can do science at all, how a community comes to represent a genuinely new dimension of the world, and how it loses information it once held. The question has a second audience, because communities of reasoning agents are now *built* as well as studied, and every such community commits, in its aggregation step, to an answer it rarely examines: what happens to the member who disagrees. Machine learning names the property at stake *open-endedness* [14, 28]; social epistemology catalogues its failure modes [23, 32]. What neither supplies is a foundation: a model of a *community* of structure-learning agents explicit enough that one can prove which aggregation dynamics preserve open-endedness and which foreclose it. This paper builds that foundation, and its central result is a negative one with constructive content: the most natural baseline—Gaussian beliefs under precision-weighted pooling—provably lacks the capacity, and even the cheapest repair the failure names—reliability inferred from surprise—restores distinctness only as a timescale; a wall appears only when the rival itself is represented, which we demonstrate at the smallest scale that exhibits it. Averaging refines a frame; it cannot grow a new one.

To change one's mind about a fact is to update a belief; to change a paradigm is to change the *frame* in which beliefs are posed—not a credence but a *structure*, a network of conditional dependencies through which information propagates. This is the phenomenon Kuhn named but could not run forward [19]: nothing in *The Structure of Scientific Revolutions* says, mechanistically, why a community holds a structure of commitments past the first disconfirming evidence, nor what finally makes the holding give way. When the chemical community gave up phlogiston it did not revise a single credence: it rewrote a network in which combustion, calcination, respiration, and the theory of acids were all load-bearing on a hub that had to go [40, 42, 43]. Phlogiston was load-bearing for too much.

Network epistemology, the dominant formalism for such communities, cannot represent this case: its agents hold a scalar credence in a fixed proposition and update it on a trust graph [16, 24, 32, 45, 75], but the phlogiston puzzle is not a failure to converge on a number—the commitments were *entangled*, and the dispute was over which entanglements to keep. Thagard's ECHO [30, 41] captures the structural intuition as undirected coherence but lacks the directed dependencies and the expand/reduce asymmetry derived below; active-inference accounts supply precision-based lock-in [1], warnings that naive aggregation degrades collective precision [47, 48], and a single-agent variational cost of mind-change [15]. What none of these does is make *belief structure itself* the object of inference: a scalar credence can never say how revising one commitment forces a re-reading of the rest—agents can herd on a number or split over it, but they cannot come to hold different *kinds* of world.

This paper lets a piece of *structure* travel, so that one agent can come to represent a commitment its peers do not. Its agents carry a directed network of commitments and learn that network by Bayesian model expansion and reduction; they are *bounded*, sampling a complex world through narrow channels, valuing some conclusions over others, and judging what they receive against their own model. Section 3 assembles this community from the field's own default choices; Section 4 opens with the two theorems under which those defaults can neither reject a conflicting source nor sustain a school—*conflict resolves into confident compromise*—and reports two results: the homogeneity effect, and its two-stage repair, reliability inferred from surprise (a timescale) and represented rivals (a wall). The discussion draws the map: what filling the representational slot bought, and what remains.

2 What a model of science must contain

Three things, at least. *Structure as the object of inference*: a scientist confronting a new domain is not given the right variables, let alone the dependencies among them; frames differ in kind, not in degree, so credences over a fixed menu cannot carry them [34, 35, 49]. Once structure is the object, exploration stops being a hand-tuned bonus [24, 32] and becomes the epistemic value of *expanding* the model to entertain a commitment not yet in it [9, 10], with Bayesian model reduction as its complement, pruning what the data abandon in closed form [11, 50]. Existing applications of that machinery run on single agents [50, 51] or define "group" as between-subject neuroimaging [52]; a *community* jointly expanding and reducing a paradigm is the open move this paper makes.

Value, because scientists care about their conclusions: a result on which a career rests is held more tightly than its evidence warrants [17, 29], and attitudes demonstrably live on networks of exactly this kind [3]. And *communication*, because science is social: beliefs travel a trust graph and must somehow be combined when they meet. Each requirement has a default formalization, so natural it is hardly ever examined. Beliefs over a network become a precision-weighted Gaussian model; value becomes a utility that tilts the prior [15]; combination becomes precision-weighted averaging, the information form of classical opinion pooling [60, 62]. Section 3 builds the community out of exactly these defaults, on purpose. The question the rest of the paper asks is whether such a community can ever do more than *refine*: whether agents built this way can grow a frame, and not merely polish the one they hold.

3 The model

We draw the system whole first—a population of agents on a trust graph—then build the single agent up from first principles, and finish by stating the dynamics as an explicit per-round loop (§3.5).

3.1 The population, and the paradigm network each agent carries

The third requirement of §2—communication—fixes the shape of the whole, so we draw the whole first. A population of N agents inhabits a shared hidden world. Agents are coupled on a trust graph: communities are blocks of a stochastic block model, with dense intra-community edges and an inter-community coupling weight (*inter*) that we sweep from total isolation to dense contact. Each agent privately carries its own *paradigm network*, a directed graph $G = (V, E, \Pi)$: nodes carry theoretical commitments x_v, edges carry inferential dependencies, and each edge bears a coupling precision π_{uv} measuring how tightly v is bound to u—from which the agent's conservatism and conviction will be derived (Eqs. 1 and 2 below). Figure 1 draws the picture.

Two kinds of things propagate on this underlying network. Beliefs and accumulated precisions are fused across trust edges in information form—an agent's own evidence and its neighbours' weighted in proportion to trust, the information-form counterpart of trust-weighted averaging in opinion dynamics [55, 60]. And—not standard—*structure* spreads the same way: a neighbour's newly awakened node arrives as a proposal, scored against the receiver's own model (§3.2)—not only a number but a piece of structure, evaluated against wiring the sender does not see.

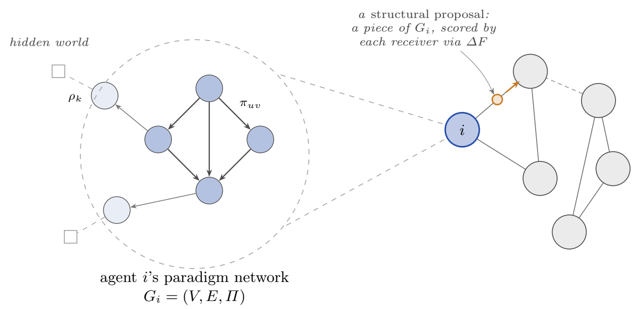

Fig. 1: The population model. Right: a trust graph, fixed at the start (stochastic block model)—dense channels within a community, one weak weight *inter* between. Left, magnified: each agent's private paradigm network $G_i = (V, E, \Pi)$, its belt reading the hidden world through channels of precision ρ_k. Along trust channels travel fused posteriors and *structural proposals*: pieces of paradigm each receiver scores against its own net by ΔF (Eq. 5).

What does one of those private networks hold? The phlogiston paradigm, at its smallest, is three commitments wired together: combustion releases phlogiston; calcination is combustion, slowed; respiration is combustion, tamed. Each commitment conditions the next, evidence for any one bears on all, and the question "what if the calx *gains* weight?" has no local answer—it tugs the whole web. A paradigm, in this paper, is that web made explicit.

The network is just a generative model whose hidden states are the commitments themselves: internal coherence is the precision-weighted graphical model $p(x \mid G) \propto \prod_{(u \to v) \in E} \psi_{uv}(x_u, x_v)^{\pi_{uv}}$—a linear-Gaussian structural equation model in the lineage of Wright's path analysis [74], edges as path coefficients and precisions as their weights. We work throughout in that regime, which puts every quantity below in closed form (Appendix A); agents are *honest*—no one misreports, so lock-in is never a dishonesty artifact—and *bounded*, minimizing variational free energy step by step [39].

Revising a commitment is not a local act: changing x_v forces everything downstream to re-equilibrate, along every chain of dependence. If A is the matrix of edge precisions, summing chains of every length gives $\sum_k A^k = (I - A)^{-1}$—a finite sum on a DAG, an object two literatures already own: the *total-effects matrix* of path analysis [58, 74], the Katz–Bonacich index of network science [59, 64, 70]. We adopt it as the *propagation operator* $\mathsf{T} = (I - A)^{-1}$: the entry $\mathsf{T}_{iw} = \pi_{i \rightsquigarrow w}$ is the total precision-weighted coupling propagated from i to w along all directed paths (Appendix A).

Conservatism is then recognized rather than posited. Place a unit of weight on every node and ask, for each commitment i, how much total weight a revision of i would disturb. That is the operator applied to the unit source,

$$\kappa \;=\; \mathsf{T}\,\mathbf{1}, \qquad \kappa_i \;=\; 1 + \sum_{w \in \mathrm{desc}(i)} \pi_{i \rightsquigarrow w}, \tag{1}$$

the precision-weighted mass of a node's descendants. In network-science terms this is a centrality (the broadcast Katz index), usually read as influence or status; we read the same number as *revision cost*—a commitment is exactly as expensive to revise as the mass that must re-equilibrate when it moves. Conservatism is a centrality the graph already determines, not a disposition assigned by hand.

What we add to this inherited machine comes in two layers. The first is contact with the world. The dense, high-precision interior of the network is the *core*; the sparse periphery in contact with observation is the *belt*, each belt node reading the world through a sensory channel of precision ρ_k—Lakatos's hard core and protective belt [66], here determined by the precision geometry rather than decreed. The core carries no direct sensory channel: an interior commitment is revised only by error propagated inward from the belt, through the same operator—costly by Eq. (1) and insulated by position, which is what makes it slow.

The second layer is value, and the operator is run twice. Scientists value some conclusions over others (§2); following the variational-cost account of belief revision [15], we attach to each node an intrinsic utility u_i and let it propagate through the *same* operator: a commitment one wants true lends value to its dependents, so the conviction field is

$$U \;=\; \mathsf{T}\,u, \qquad U_i \;=\; u_i + \sum_{w\in \mathrm{desc}(i)} \pi_{i\rightsquigarrow w}\, u_w. \tag{2}$$

In the operator's path-analytic reading, U is the total effect of wanting things true. Where the inherited models propagate a single source, a paradigm carries two: $\mathsf{T}\mathbf{1}$ and $\mathsf{T}u$ are linearly independent unless $u \propto \mathbf{1}$, so the two fields need not align—a commitment can be central yet value-neutral, cheap yet strongly held (Fig. 4, Appendix A).

3.2 Motivated revision, structure learning, and forgetting

The bounded agent revises its belief $q(x)$ over the whole network by the motivated trade of Hyland and Albarracin [15]: the variational cost of changing its mind—divergence from the prior its own Bayes net encodes—against accuracy to the evidence and the value of where the commitments land,

$$\begin{aligned} \mathcal{F}[q] \;&=\; \underbrace{D_{\mathrm{KL}}[q(x)\,\|\,p(x\mid G)]}_{\text{mind-change cost}} \;-\; \underbrace{\mathbb{E}_q[\log p(o\mid x,G)]}_{\text{accuracy}} \;-\; \lambda\, \underbrace{\mathbb{E}_q\big[U^{\top}x\big]}_{\text{value}} \\ \Longrightarrow \qquad & q_\lambda(x) \;\propto\; p(x\mid o,G)\, e^{\lambda\, U^{\top}x}, \end{aligned} \tag{3}$$

with λ a single temperature governing how far value may bend the evidence. The KL is anchored to the *network prior*: mind-change is priced from where the mind currently stands, in the metric of the paradigm's own precision (Appendix A, Eq. 8), and the minimizer is classical—the exponential tilt KL-regularized utility maximization always yields [72, 73]. What is ours is the source of the tilt, a conviction field propagated through the paradigm's own operator. In the linear-Gaussian regime the tilt is a pure mean shift, displacing the commitments by $\lambda \Sigma U$ and leaving the covariance untouched. What opposes the displacement is evidence, also through the mean: a disconfirming channel k with sensory precision ρ_k and read-out H_k pulls the mean back with gain $\propto \rho_k H_k \Sigma U$—so as conviction drives $\rho_k \to 0$, the value-displaced commitment stands unrefuted. The fraction of disconfirming sensory precision so silenced,

$$\gamma \;=\; \frac{\sum_{k\in\mathrm{dis}} \rho_k \nVdash[\, \rho_k \text{ gated by the core}\,]}{\sum_{k\in\mathrm{dis}} \rho_k}, \tag{4}$$

is the *lock-in parameter*. Nothing is silenced by hand: each round channel k carries weight $w_k = \exp(-g\,|U \cdot H_k|)$—the more directly a channel bears on valued commitments, the less the agent listens, with g setting how strongly caring becomes not-listening. Note what this mechanism *is*: a displacement of where beliefs come to rest—under the forgetting below, a *saturating* one while evidence keeps arriving. A bias, not a resistance; the results make that distinction load-bearing.

Structure learning revises G itself—the first requirement of §2—by two asymmetric operations [10, 26], the formal counterpart of normal versus revolutionary science. *Reduction* is the exact half: each edge carries an automatic-relevance prior precision [20, 31], and Bayesian model reduction [8, 11] scores any pruning in closed form from the already-inverted posterior,

$$\frac{p(o \mid \tilde{M})}{p(o \mid M)} = \mathbb{E}_{p(\theta|o)}\left[\frac{\tilde{p}(\theta)}{p(\theta)}\right], \tag{5}$$

the Savage–Dickey ratio in general form: one inversion scores every candidate reduced structure without re-touching the data.

Expansion reaches what reduction cannot—a commitment no one has represented—by wiring a new node in with free priors and letting reduction prune it back to the couplings the data support [10]. Expansion requires a real inversion, and the cost asymmetry is why a field reorganizes its periphery continually and its centre only rarely. Candidates arrive at a Poisson rate—one cannot plan toward a dimension one has not yet conceived [27]—and are kept only if the model log Bayes factor favours the wired-in node (Appendix D).

Finally, *forgetting*: accumulated precision relaxes toward the prior, $\Pi \leftarrow \Pi_0 + \omega(\Pi - \Pi_0)$, $\omega \in (0, 1]$ [36, 38]—one forgets at the rate one believes the world's structure is moving [37]. Without it $\kappa = \mathsf{T1}$ can only grow and lock-in is automatic rather than earned; with it, the value tilt saturates at $\lambda U/(1 - \omega)$ against also-saturated per-step evidence, so conviction holds only above a threshold it must clear.

3.3 Reliability as a latent

One default remains, so quiet it is rarely stated: every precision above is *known*—the sensory ρ_k, the trust weights of the fusion step—constants, blind to what the source actually says. The results open with what that costs (two theorems; §4), and the conflict-resolution literature has long owned the repair [6]: stop treating a source's precision as a constant and *infer* it. Carried as a scale mixture, each channel's noise precision becomes a latent with its own prior, and its posterior expectation, given the standardized one-step surprise z_k^2 (the squared residual of observation k against the agent's own prediction, in units of its predictive variance), is the classic Student-t weight

$$\eta_k \;=\; \frac{\nu+1}{\nu+z_k^2}, \tag{6}$$

applied multiplicatively to the channel's evidence deposit. The rule is one sentence: *trust what does not keep surprising you.* A channel reporting on-prediction ($z^2 \sim 1$) keeps full weight; a gross outlier is discounted as ν/z^2; the single dial ν sets how quickly surprise becomes doubt, and $\nu \to \infty$ recovers the Gaussian defaults exactly. The same functional gates trust *between* agents, $\eta_{ij} = (\nu_s+1)/(\nu_s+z_{ij}^2)$, with z_{ij}^2 the calibrated disagreement between i's and j's posteriors—and because z^2 is just a calibrated gap, the gate can be pointed at *opinions* (the posterior means) or at *wiring*, the contested coupling entries of Π itself (definitions in Appendix C). Note what the gate is not: it adds no objective and no memory, every weight stays strictly positive, and so it creates no new fixed point. Both gates default to off, reproducing the baseline byte-identically.

3.4 Represented rivals

The gate infers how much to trust a source; it cannot represent the *alternative* a source argues for. The last mechanism supplies that slot, at the smallest scale that makes it testable. Each agent carries K candidate wirings (Π_k, h_k) and a posterior over a model index—$q_i(x,m)$ rather than $q_i(x)$—and each step every candidate is scored by its prequential predictive density *before* the shared deposit lands, so the structure-discriminating evidence the Fisher deposit erases accumulates in the log-weights L_{ik}. Communication changes accordingly: parameter beliefs fuse only *within* a frame (candidate k with neighbours' candidate k, never averaged across frames), and what travels *between* frames is the hypothesis itself, by damped log-linear pooling of the log-weights, $L \leftarrow (1-\alpha_m)L + \alpha_m WL$. The damping is the point: it gives the pooled log-odds a genuine fixed point at a constant gap set by each camp's own evidence source ($\sim s/2\alpha_m$ nats for two symmetric camps), where the gates of §3.3 leave consensus as the only fixed point. $K=1$ reduces exactly to the baseline's fuse-then-observe, so the comparisons of §4 run all three architectures on byte-identical observation streams (definitions in Appendix C).

3.5 The simulation at a glance

The sweep of Appendix B instantiates the phlogiston world: a hub-structured paradigm net in which one core commitment is load-bearing for combustion, calcination and respiration; at a fixed step the world flips (the calx is heavier—the gravimetric anomaly), and the truth thereafter contains a dimension (oxygen) in *no agent's network*. One round of the simulation, for every agent:

1. **Observe.** The world emits observations at the belt; disconfirming channels carry the conviction-gate weight w_k (§3.2) and, with the reliability gate on, the inferred η_k (Eq. 6).
2. **Infer.** Exact linear-Gaussian update of the value-tilted posterior (Eq. 3); precision accumulates.
3. **Crisis check (reduction).** Candidate prunes are scored by $\Delta F + \lambda_i \, \Delta U$, both terms closed-form from the Savage–Dickey reduced posterior (Eq. 5); both protective belt edges clearing the bar is a *crisis*, and the hub is released (Appendix D).
4. **Discovery check (expansion).** When a Poisson proposal arrives, its direction is read from the post-reduction residual's leading eigenvector, and the wake is kept iff the model log Bayes factor is positive (Appendix D).
5. **Fuse.** Posteriors and structural edits pool across the trust graph, weighted by the intra/inter coupling and, with the social gate on, by η_{ij} (§3.3); pooling all dimensions vs. only those jointly represented is a protocol choice with consequences (Appendix B).
6. **Forget.** Accumulated precision decays by ω (§3.2).

Crisis, revolution and discovery are thus not stipulated events but threshold crossings of the model's own evidence; Appendix D derives both checks from the single tilted objective and lists explicitly what remains a modelling commitment. Table 1 (Appendix B) lists the dials swept; everything else is held fixed, and all code is seeded and ships its own null and positive controls.

One thing deserves emphasis before the results: every baseline ingredient above is its field's default—Kalman-filter belief updating, the standard formalization of motivated reasoning, classical opinion pooling in information form, textbook Bayesian model comparison. Nothing exotic has been smuggled in; the results test what these defaults can do *together*.

4 Results

All results are from the population of §3.1, run by the loop of §3.5. The main text reports two experiments, each chosen to isolate one claim at the smallest scale that exhibits it; the full Kuhnian-cycle sweep on the phlogiston world—crisis, reduction, and discovery as threshold crossings of the agents' own evidence, none stipulated—is consistent with both and carried in Appendix B. One signature organizes everything we find: *conflict resolves into confident compromise*. It is not an accident of parameters. Two classical theorems, one per level of the model, say it could not have come out otherwise.

Pooling is a contraction. The fusion step is precision-weighted averaging on a connected trust graph—DeGroot dynamics in information form [60]—and its weights never depend on the *content* of what is pooled. Any such step contracts the spread of beliefs geometrically toward consensus: disagreement can only decay, never regenerate. A persistent school of thought would require an agent to weigh a message by its fit to the agent's own structure, and no quantity in the loop of §3.5 reads that fit; the hazard is recognized within active inference itself [2]. *Gaussian conflict resolves by compromise.* Within an agent the same impossibility holds one level down, and it has been a theorem for fifty years: a posterior built from light-tailed sources responds to conflict between them by confident compromise—it can never reject a conflicting source, however extreme [4, 5]—and its precision *grows* while doing so [6], so contradiction makes a Gaussian agent more certain, never less. There is no slot in (Π, h) for being torn: crisis as suspended confidence requires tails the regime does not have.

The two theorems make this community a *null regime*—the simplest faithful model of a believing, communicating community, in which Kuhnian phenomena provably cannot arise. But they are qualitative: what they do not say is where the boundary sits, or how sharply it cuts. We measure both, as two results: the homogenization the theorems demand (§4.1; the full characterization in Appendix B), and what happens when the one quantity the theorems hold fixed—each source's reliability—is inferred instead (§4.2; Appendix C). The nets are hand-wired to expose the geometry, not fitted to data: nothing here is evidence about chemistry, only about how a structured, bounded, valued mind changes.

4.1 Result 1: the homogeneity effect

The experiment is simple. The world is built so that its data admit two equally good readings—two different ways of wiring the same commitments into a paradigm network. We split the population into two communities and give them different values, so each community attends to the observations that favour one of the two wirings; we then measure the *structure distance* between the networks the agents actually learn—small when two agents wire the world the same way, large when they wire it differently. The question is whether one population can come to hold both readings.

Kept apart, it can (Fig. 2, blue): each community settles on its own wiring, and the distance between them grows large and stays. Connected, pooling every round (red), the divergence never forms: each fusion step averages the two half-formed wirings into a compromise network neither community's evidence supports—a tenth of the disconnected distance. This is Result 1, the *homogeneity effect*: under default pooling a connected population holds one network, whatever diversity its members were learning, because the contraction theorem leaves the dynamics nowhere else to go.

The phlogiston sweep (Appendix B) finds the same effect at every level of the model. *Conviction does not protect:* valuing a conclusion shifts where beliefs settle, never when they yield [15]. *Isolation is the only protection, and it is fragile:* at *inter*$=0.005$—trust toward outsiders two hundred times weaker than within—92% of a dogmatic community converts, because an agent can silence its

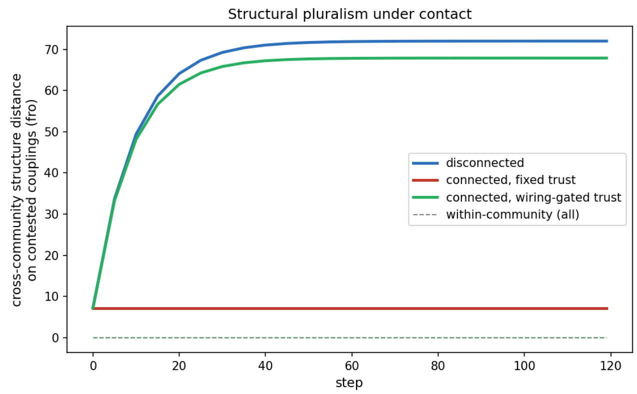

Fig. 2: The paper's two results in one experiment (a structurally underdetermined world; cross-community distance on the contested couplings, mean ± s.d., 5 seeds; dashed: within-community, ∼0 throughout). Apart (blue): two crisply different wirings. Connected, default fusion (red): the pluralism never forms—10% of the disconnected divergence, the homogeneity effect (Result 1). Connected, wiring-reading trust (green, §3.3): 94% survives full contact (Result 2).

own instruments but not its neighbours'. *Discovery does not survive company:* a concept woken by a single agent is averaged away by peers who treat "has no such variable" as "certainly zero." *No added mechanism escapes:* adequacy-driven exploration, social excitation, audited curiosity, and value accretion each re-time the passage to consensus or synchronize the population for it. The effect even has a one-line algebraic face: $|\,\text{mean}_i\, \Pi_i\,| \le \max_i |\Pi_i|$ entrywise, so an average can never *create* an edge absent from every individual—whatever a collective knows beyond its members lives in the *union* of their networks, never in their average.

4.2 Result 2: distinctness returns—gates buy time, represented rivals buy walls

The same setups, the same evidence; what changes is only what the agents *represent*. First, switch on the gate of §3.3. Within an agent, one channel of a three-commitment chain turns persistently contradictory at step 30: the Gaussian baseline does what Theorem 2 says it must—it settles confidently between the channel's report and its own couplings, variance falling monotonically, the conflict leaving *no trace*—while the gated agent infers the channel unreliable ($\eta \to 0.16$) and its uncertainty does what no light-tailed posterior can: it *rises* while evidence accumulates, peaking before the belief reorganizes around the discounted channel (Fig. 8, Appendix C). There is now a slot in the belief state for being torn—crisis as suspended confidence, rendered as a posterior.

Between communities, point the same functional at the *wiring*—the contested couplings of Π—and the collapse of Fig. 2 reverses (green): 94% of the disconnected divergence survives full contact where fixed trust keeps 10%. What the gate *reads* decides: pointed at *opinions* it never fires here (10.2%), because both camps fit the same blended data and agree in their means—a pluralism that lives in wiring is invisible to a trust that reads opinions.

But the gate is memoryless and its weights stay strictly positive, so the contraction theorem still owns $t \to \infty$: every school it sustains is metastable. Figure 3 makes the hierarchy visible on one long run—two camps in full contact, each running only its own commitment's experiment (theory-laden evidence, the Kuhnian condition), three architectures on byte-identical streams. Plain fusion collapses within a few steps: the theorem. The trust gate holds the gap for hundreds of steps, then visibly decays: time, bought and spent. The rival layer of §3.4 holds a flat gap to $t = 1500$, each camp internally certain of its own theory: a wall—a fixed point of the damped hypothesis-pooling dynamics, not a slow leak (Appendix C).

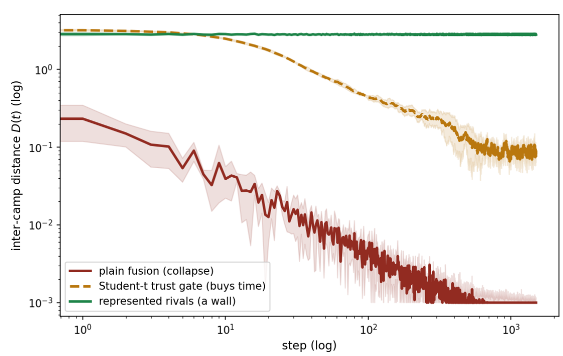

Fig. 3: Same setups, same evidence—only the representation differs (two camps, full contact, theory-laden streams; inter-camp distance $D(t)$, mean ± s.d., 5 seeds, log–log). Plain fusion collapses (the theorem); the Student-t gate decays (time, bought and spent); represented rivals hold (a wall). Each architecture runs its own best memory setting (Appendix C).

Boundaries, stated plainly. The demonstrations are deliberately small—one agent, $N = 60$, then $N = 16$, on hand-wired worlds—existence proofs that the wall was representational, not sweeps of social topologies. The rival mechanism is *given* its K candidates, and anchored forgetting on the candidate nets is part of the mechanism—keeping a rival represented means not letting accumulated evidence drown it. What it does not yet do is named: under log-linear pooling a lone discoverer's candidate is majority-swamped, and no agent yet *acts* on where its candidates disagree—the crucial experiment. Controls: a world that never shifts produces no doubt; gates and rivals off reproduce the null regime byte-identically; no bar is tuned (Appendix C).

5 Discussion

Taken together, the results say the homogenization is a failure of *representation*, not of tuning. Result 1 is the theorems made visible: a community built from its field's defaults is the formalism's *normal-science* limit, and nothing in it entrenches, diverges, or breaks. Result 2 shows the wall stands exactly where the theorems put it: make reliability a latent and an agent can be surprised into doubt, a community can hold two wirings under contact—yet every protection the gate wins is a clock; represent the rival itself and a wall finally stands.

Trust as inferred precision—now measured, and bounded. The gate is the conflict-resolution literature's canonical repair [6]: within an agent it buys crisis rendered as a posterior; between agents, stable schools on a connected graph holding shared data. But everything it buys, it buys as *time*: the gate is memoryless and its weights stay strictly positive, so consensus remains the only place the dynamics can rest.

The missing slot is representational—and fillable. There is no coordinate in (Π, h) whose value can mean *torn between two hypotheses*: conflict arrives as growing confidence, and fusion erases who disagreed about what. This impoverishment is by representation, not parameters—why the gate could only slow it. The rival layer of §3.4 is that diagnosis made executable, and the wall of Fig. 3 is what filling the slot buys: multiple structural hypotheses per agent (a genuinely torn agent), divergence between live candidates as a computable state, and communication of hypotheses—fusion within frames, never averaging across them. What remains open is named: a lone discoverer's candidate is majority-swamped under log-linear pooling, and no agent yet *chooses* the experiment on which its candidates disagree most. Gates buy time; represented rivals buy walls.

The relation to Kuhn is now allegory-free: the regime *holds* normal science (the derived core and belt of §3.1). Of the revolutionary remainder, inferred reliability *supplies* crisis as suspended confidence and renders incommensurability [65] a timescale, while represented rivals make it a *state*: two camps internally certain of different wirings at a fixed point, the gestalt switch given its slot (a $q(m)$ that flips on prequential evidence).

Any community whose aggregation step averages rather than gates inherits the regime's signature, whatever the sophistication of its members: a population of structure learners is only as *open-ended* as its pooling permits—and as what its pooling *reads* (§4.2)—a property of the protocol, designed in rather than an emergent grace of scale. The deepest exit lies beyond the model class—a genuinely new *instrument* enlarges the likelihood's own support, and one cannot place a posterior over the unconceived [27]. What a community controls is how hard it looks—and, above all, what its pooling does to the member who finds it first.

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A The operator, the tilt, and the Schur residue

Linear-Gaussian construction. The couplings form a structural equation model $x = Ax + \zeta$, $\zeta \sim \mathcal{N}(0, D^{-1})$ with A the weighted DAG adjacency [18]; the choice is for tractability, and a nonlinear coupling would replace T by a local linearization. The propagation operator is the resolvent

$$\mathsf{T} = (I - A)^{-1} = \sum_{k \geq 0} A^k, \qquad \mathsf{T}_{iw} = \sum_{\text{paths } i \rightsquigarrow w} \prod \pi \;=\; \pi_{i \rightsquigarrow w}, \tag{7}$$

the Neumann series finite because a DAG makes A nilpotent. We realise T on the coupling magnitudes $|A|$ so downstream mass does not cancel when a child opposes a parent; the signed A builds the precision matrix. We index T so that κ_i sums row i; under the SEM convention the released code stores the transpose and computes the same fields as column-sums of $\mathsf{T}^\top$, so a reader reproducing $\mathsf{T}\mathbf{1}$ literally against the code's array must transpose first.

The value tilt. With honest posterior $p(x \mid o, G) = \mathcal{N}(\mu, \Sigma)$, the tilted posterior of Eq. (3) is a pure mean shift, $q_\lambda(x) = \mathcal{N}(\mu + \lambda \Sigma U, \, \Sigma)$, with cumulant-generating function $Z(\lambda) = \lambda \, U^\top \mu + \frac{1}{2}\lambda^2 \, U^\top \Sigma U$. Note that gating a channel ($\rho_k \to 0$) *raises* the posterior variance and hence Z''; lock-in lives in the vanishing mean-update gain $\rho_k H_k \Sigma U$, not in the curvature.

The mind-change cost of Eq. (3) is anchored to the network prior $p(x \mid G) = \mathcal{N}(\mu_0, \Pi_0^{-1})$: the revision q_λ pays

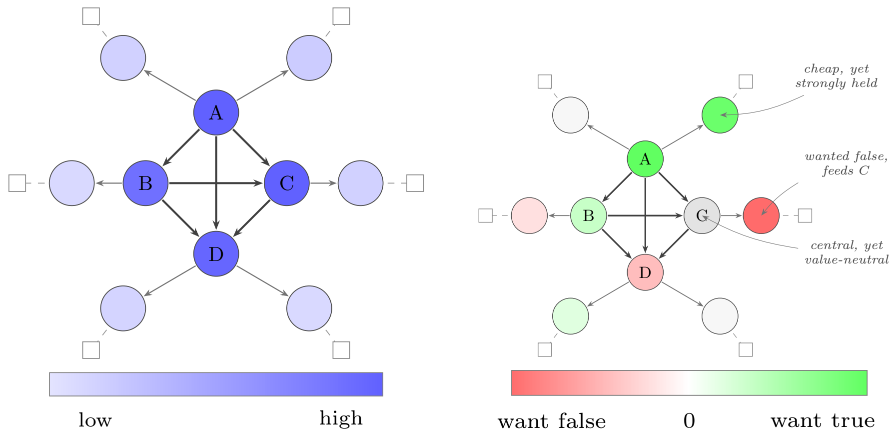
$$D_{\mathrm{KL}}[q_\lambda \,\|\, p(x \mid G)] = \tfrac{1}{2}\, \delta^\top \Pi_0 \, \delta + \tfrac{1}{2}\big[\mathrm{tr}(\Pi_0 \Sigma) - \log\det(\Pi_0 \Sigma) - d\big], \tag{8}$$

with $\delta = \mu + \lambda \Sigma U - \mu_0$ and $d = \dim x$: the evidence-driven and value-driven displacements are priced *together*, in the prior precision Π_0—exactly the off-diagonal mass the carry-over field $\kappa = \mathsf{T}\mathbf{1}$ summarizes. The smaller quantity $D_{\mathrm{KL}}[q_\lambda \,\|\, p(x \mid o, G)] = \frac{1}{2}\lambda^2 \, U^\top \Sigma U$ is the fidelity gap to the honest posterior, not the cost of changing one's mind: it would vanish for an agent that followed the evidence exactly, however far the evidence moved it.

The Schur residue. Marginalizing a hub condenses its influence into new edges among its neighbours: for a common-cause structure over (x_1, x_2, b),

$$\Pi = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{\text{marginalize } b} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \tfrac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1 \;\, 1) = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{pmatrix}, \tag{9}$$

a direct coupling induced where the joint had a zero. Conditioning on a shared *effect* is the dual operation: two commitments that jointly feed one observation are a priori independent, and clamping that observation anti-couples them—explaining away, run on the precision matrix. Both operations fill in the off-diagonal coupling the carry-over cost is built from, and Fig. 5 draws both on the smallest nets that exhibit them.



(a) Revision cost κ_i (conservatism) (b) Intrinsic utility u_i (conviction)

Fig. 4: Two fields on one paradigm (schematic). Both panels carry the *same* network—a densely coupled core $\{A, B, C, D\}$ and a sparse belt $\{b_1, \ldots, b_6\}$ in contact with observation. a Revision cost is the carry-over of mind-change, $\kappa = \mathsf{T}\mathbf{1}$ (Eq. 1): a node's cost is the precision-weighted mass of its descendants, a clean radial gradient monotone in centrality. b Intrinsic utility is signed—beliefs one wants true (green), wants false (red), or is neutral toward (white)—and the conviction field $U = \mathsf{T}u$ does not align with κ: C is central yet value-neutral; b_2 is cheap yet strongly held; b_4, which feeds C, is a conclusion the community wants false. The two fields are $\mathsf{T}\mathbf{1}$ and $\mathsf{T}u$, the same operator on two sources, and they are independent vectors.

B The null regime, measured

This appendix carries the quantitative characterization of the default (fixed-reliability) community summarized in §4.1: the full Kuhnian-cycle sweep on the phlogiston world—normal science, the gravimetric anomaly, crisis, reduction, discovery of oxygen, every phase a threshold crossing of the agents' own evidence, none stipulated—over the dials of Table 1, spanning the baseline and four added mechanisms (Table 2).

Only total isolation protects a superseded paradigm. The setup pits a dogmatic community—high conviction, whose agents gate their own disconfirming channels (§3.2)—against an open vanguard already tracking the anomaly, and dials the inter-community trust weight *inter* from 0 (sealed off) to dense contact. Sealed off, the gating protects: under strong gating ($s = 10$) only 19% ever convert. But at *inter*$= 0.005$, 92% convert to oxygen, however strong the gate and however dogmatic the agents, and the transition is a sharp boundary, not a gradual fade (Fig. 6). A control pins down what does the stalling: keep conviction exactly as high but switch the self-gating off, and the community revolts in full

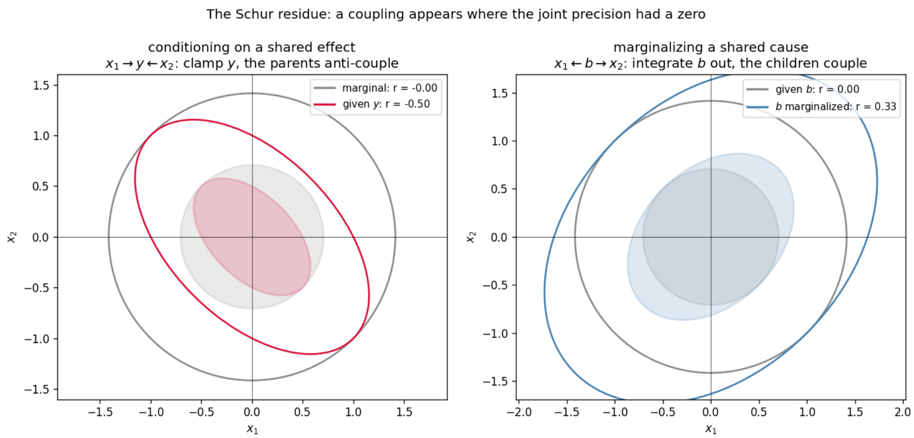


Fig. 5: The Schur residue, computed on the smallest nets that show it. *Left:* conditioning on a shared effect ($x_1 \to y \leftarrow x_2$): the parents are a priori independent (grey, $r = 0$); clamping their common descendant y manufactures an anti-correlation (crimson, $r = -0.50$)—explaining away. *Right:* marginalizing a shared cause (the matrix of Eq. 9): the children are independent given b (grey); integrating b out leaves them coupled (blue, $r = +0.33$). In both cases a coupling appears where the joint precision had a zero—the off-diagonal mass the carry-over cost $\kappa = \mathsf{T1}$ is built from.

Dial	Symbol	What it controls
Social coupling	*inter*	inter-community trust weight (0 = isolation)
Vanguard fraction	—	fraction of agents in the open (low-λ) community
Conviction	λ_d	dogmatic community's value tilt / prune protection
Gate scaling	s	strength of the conviction gate on disconfirming channels
Proposal rate	r (Poisson)	arrival rate of structural proposals
Forgetting	$\omega \in (0, 1]$	decay of accumulated evidence toward the prior
Observation noise	σ_o	sensory noise at the belt
Channel reliability	ν	Student-t gate on the agent's own channels (off = Gaussian)
Social trust	ν_s	Student-t gate on the fusion weights (off = fixed trust)
Fusion protocol	—	pool all dimensions (delta) vs. jointly represented (abstention)
Closed-loop mechanisms	$\beta, \tau, \kappa_r, \lambda_c, \epsilon$	adequacy-driven rate, social excitation, audit credit, value accretion (Table 2)

Table 1: The dials swept in the experiments; all other quantities are held fixed. The last two rows are protocol and mechanism variations layered on the baseline; each defaults to "off" (constant rate, one-shot accept, delta pooling, static u, both reliability gates off), reproducing the baseline model exactly.

(revolted fraction 1.0 versus 0.0, $\gamma \to 1$). The tilt never stalls anyone (Eq. 4); its one causal route to lock-in runs through the gate, and the gate loses to the faintest connection, because a neighbour's evidence arrives weighted by trust, not by the receiver's conviction. Motivated reasoning in this regime is *wishful*, not *dogmatic*: a bias lives in the potential h; an entrenchment would have to live in the precision Π—which is exactly where the inferred-reliability gate of §3.3 puts it.

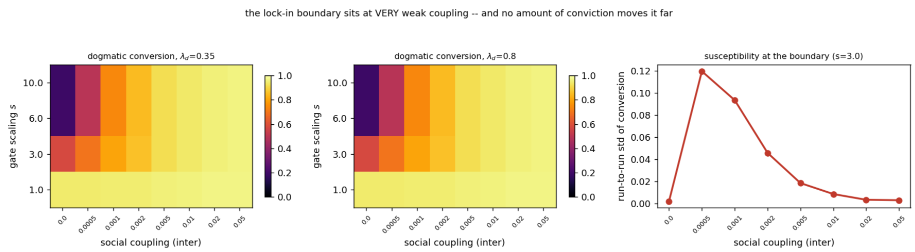


Fig. 6: Only isolation protects a superseded paradigm (phlogiston sweep, fixed trust). *Left, centre:* the fraction of the dogmatic community that converts to oxygen, as a function of contact (*inter*) and of how strongly conviction gates disconfirming channels (s), at two conviction levels ($\lambda_d = 0.35$ and 0.8). Past *inter* ≈ 0.005 nearly everyone converts, and doubling conviction barely changes the picture. *Right:* the run-to-run spread spikes in a narrow band around *inter* $\approx 5 \times 10^{-4}$: a sharp boundary, not a fade.

Pooling kills the lone discoverer—by protocol, not by principle. Discovery means waking a node—oxygen—that no agent's network yet contains; candidates arrive at random (Poisson rate r, §3.2), and the fraction that discovers follows the waiting-time law $1 - (1 - r)^T$. What the rate governs is survival: under the baseline fusion, a discovery made by one agent at a time is averaged away every time, because delta pooling reads *has no such variable*—a fact about an agent's model class—as *maximally certain it is zero*, a belief within its support [47, 62]. The corrected protocol is dimension-aware *abstention* pooling: each entry of the precision matrix is pooled only over the neighbours that represent it. The consequence is a wholesale relocation of the survival threshold (Fig. 7): under delta pooling staggered survival is zero at every rate; under abstention pooling it tracks the discovery fraction itself (0.39/0.64/0.86 at $r_0 = 0.005/0.01/0.02$).

Every escape goes around the averaging, never through it. The sweep's remaining mechanisms sharpen the same conclusion (Table 2). None of them lets two connected communities holding the same data *remain* in disagreement: each either re-times the passage through the consensus bottleneck or synchronizes the population for it.

Robustness. The arc is robust across forgetting and observation noise ($\omega \in [0.9, 1.0]$, $\sigma_o \in [0.25, 1.0]$): crisis and reduction survive everywhere—including the no-forgetting wall $\omega = 1$, where only discovery thins: a community that never

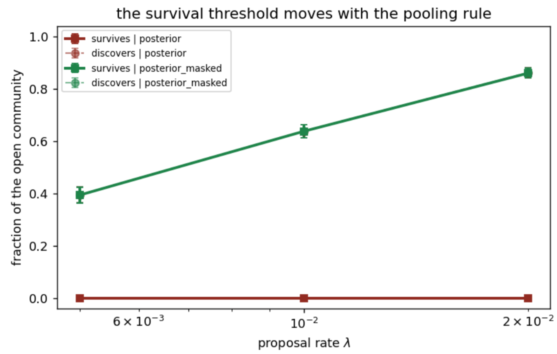


Fig. 7: The survival threshold moves with the pooling protocol (32 seeds per cell). Solid: fraction of the open community whose concept survives; dashed: fraction that discovers at all. Under delta pooling (red) a pinned slot is a confident "no such thing" and staggered survival is zero at every rate; under abstention pooling (green) survival tracks discovery. Incommensurability as a population-level force is a property of the protocol, not of the concepts.

Escape	What it changes	Measured effect	Verdict
Adequacy-inferred rate	*when* the community looks	crisis-to-discovery 55 → 29 steps; matched by a constant rate of equal mean	re-times the passage
Social excitation (Hawkes)	how nearly *together* it looks	waking dispersion 64 → 2 steps; staggered survival 0 → 0.88–0.95	synchrony outruns the averaging
Audited wake-then-prune	whether curiosity is *safe*	null world: 100% of wakes re-pinned, exactly	keeps the search honest
Value accretion	conviction grown from *inside*	conversion delayed ∼ 2.2×, never prevented	buys time, not immunity

Table 2: Four mechanisms layered on the fixed-reliability baseline, none of which opens the consensus engine.

forgets still sees its paradigm die, but loses the capacity to grow the successor. Each experiment ships its own null and positive controls—a world that never shifts produces neither crisis nor discovery; the gated, isolated community stays locked; Poisson discovery follows the waiting-time law—and no acceptance bar is tuned anywhere: the wake's sole accept test is the model log Bayes factor (Appendix D); whatever the derivations there do not fix is named there as a commitment.

C Inferred reliability and represented rivals: definitions and experiment details

The three reads of one functional. Every gate in the paper is the Student-t scale-mixture weight $\eta = (\nu + 1)/(\nu + z^2)$—the posterior expectation of a source's latent noise scale given its standardized squared surprise z^2 (one E-step of the EM view of t-regression)—applied to three different surprises. *Channels:* $z_k^2 = (o_k - (H\mu)_k)^2/\mathrm{predvar}_k$ with $\mathrm{predvar}_k = \sigma_o^2 + (H\Pi^{-1}H^\top)_{kk}$, the agent's own surprise at observation k before depositing it; η_k multiplies the channel's deposit, composing with the conviction gate of §3.2. *Opinions:* $z_{ij}^2 = \sum_v (\mu_j[v] - \mu_i[v])^2/(\Pi_i[v,v]^{-1} + \Pi_j[v,v]^{-1})$, the calibrated disagreement of two posteriors summed over the measured nodes (summed, not averaged: a localized heresy must be able to sever trust); η_{ij} multiplies the fusion weights, whose row normalization absorbs the overall scale. *Wiring:* the same sum taken over contested coupling entries $x = \Pi[a,c]$, normalized by their pairwise symmetric scale $\max((x_i^2 + x_j^2)/2, \epsilon^2)$—bounded per coupling, and invariant to the global precision growth the forgetting equilibrium produces, so it reads *how* two agents wire the world, not how much evidence they hold.

Represented rivals: definitions. Each agent carries K candidate nets (Π_{ik}, h_{ik}) and log-weights L_{ik} over a model index. Per step, every candidate is scored by its prequential predictive density *before* the shared deposit lands, $\log p(o_t \mid o_{<t}, M_k)$, so structure-discriminating evidence accumulates in L rather than being erased by the Fisher deposit (which never sees the observation's value). Fusion is hypothesis-aligned—candidate k pools only with neighbours' candidate k—and the log-weights pool log-linearly with damping α_m: $L \leftarrow (1 - \alpha_m)L + \alpha_m WL$. Deviations of the pooled log-odds from the population mean converge to $(I - W)^+(s - \bar{s})/\alpha_m$, a constant gap of $\sim s/2\alpha_m$ nats for two symmetric camps: a genuine fixed point—the wall of Fig. 3—whose height the early-measured per-step source predicts conservatively (realized 110 nats at the horizon, tail still growing at +0.016 nats/step). Anchored forgetting on the candidate nets, $\Pi_{ik} \leftarrow \Pi_{0k} + \omega(\Pi_{ik} - \Pi_{0k})$, is part of the mechanism: without it every candidate's precision grows without bound and their predictive densities converge. $K = 1$ reduces exactly to plain fuse-then-observe, which is how the Gaussian baselines run on byte-identical observation streams; each architecture runs its own best memory setting.

The experiments. Table 3 lists the gated configurations; all are seeded, assertion-checked at run time, and write their summary statistics next to their figures. The rows below the rule are population-level runs of the same gates through the full Kuhnian cycle—denial and rehabilitation of doubted channels, schism thresholds in accumulated confidence, self-organized and then self-undone isolation—reported here rather than in the main text because each is a further reading of the same two results: homogenization delayed, never escaped.

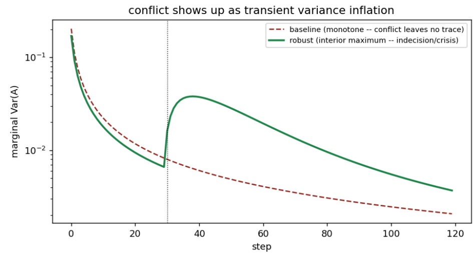


Fig. 8: Conflict as suspended confidence (§4.2). A channel turns contradictory at step 30. The Gaussian baseline (dashed) grows monotonically more certain—Theorem 2's confident compromise; the gated agent's variance (solid) shows an interior maximum: doubt rises while data accumulate, then resolves.

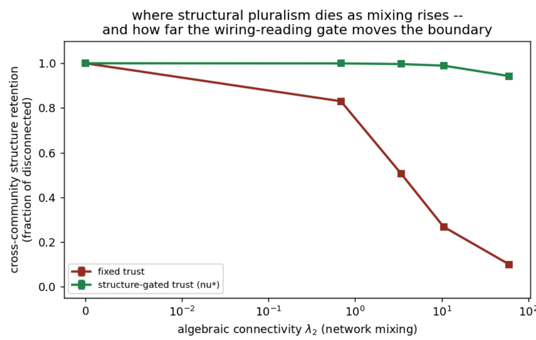


Fig. 9: Where structural pluralism dies as mixing rises. x-axis: algebraic connectivity of the contact graph; y-axis: surviving cross-community structure (fraction of the disconnected benchmark). Fixed trust dilutes both wirings toward one compromise net as mixing rises; the wiring-reading gate preserves the two learned structures even under dense contact. The opinion-reading gate tracks the fixed-trust curve: the camps agree in means, so it never fires.

Scope, stated plainly. Three commitments bound the results. (i) Both gates are *memoryless*: η_k and η_{ij} are instantaneous functions of current surprise, so every school and schism reported is metastability relative to a horizon, not a new fixed point—the contraction theorem owns $t \to \infty$, and the lock-in rerun shows the trade directly: a harsher gate (ν_s=0.2) holds the boundary longer but leaves its own undoing unfinished at the same horizon, because the lock and the merge run on the same clock. (ii) The schism control is normalized by the pre-contact stance gap, since one uniform fusion step collapses the fixed-trust camps before the first logged sample. (iii) Structure-as-precision-pattern is attention-driven in the linear-Gaussian class (the Fisher deposit $H^\top \mathrm{diag}(w) H/\sigma^2$ never sees the observation's value): a camp committed to a wiring the data do not support

Experiment	Setup	Headline
Conflict crisis (§4.2)	1 agent, chain $A{\to}B{\to}C$, channel A contradictory from $t{=}30$; $\nu{=}4$, $T{=}120$	$\eta_A \to 0.16$; interior variance maximum at $t{=}38$; strain ratio 5.3× on the violated edge
Divergent wiring (§4.2)	$N{=}60$, structurally underdetermined world, two attention camps, $T{=}120$, 5 seeds	retention: fixed 10%, opinion-gate 10.2%, wiring-gate 94%; synergy = 0
Rival lock-on (§3.4)	1 agent, $K{=}2$ wirings differing in nothing but structure, $T{=}400$	$q(M_1) \to 1.0$ at 0.13 nats/step; the Gaussian pool's half-strength blur frozen ($< 10^{-3}$ drift)
The wall (§4.2)	$N{=}16$, two camps, full contact, theory-laden channels, $T{=}1500$, 5 seeds	end $D(t)$: plain 2.5×10^{-4}, gated 0.075, rival 2.79; $q_{\text{own}} = 1.0$ both camps
Targeted reduction	1 agent, chain with one world-violated coupling, $T{=}80$	violated edge top-strain at 100% of steps; ΔF verdicts correct from step 1
Crisis trace	$N{=}20$, flip at $t{=}40$, $\nu{=}2$, $T{=}800$	denial $\bar{\eta}: 0.85 \to 0.17$; conversion stalled 4.3×; rehabilitation to 1.27; agreement channels $>$ 0.85 throughout
Gated camps	$N{=}20$, complete graph, opposed camps, $\nu_s{=}0.5$, $T{=}80$	uniform fusion: 0.1% of disagreement survives; gated: 48%; cross/within trust 0.8%
Schism threshold	ditto, sweeping prior confidence at contact	fixed trust $\leq$ 0.2% everywhere; gated $<$ 1% below the threshold, 17/48/76% above
Lock-in rerun	$N{=}80$, full cycle, $\nu_s{=}0.5$, $T{=}180$, *inter* swept	$t_{1/2}$: 78 → 146–170 at the boundary; conversion 0.54–0.73 vs 0.92; trust $<$ 1% then $>$ 90%

Table 3: The gated experiments: configurations and headline numbers. Above the rule, the two experiments of §4.2; below it, the population-level readings of the same gates.

consolidates it exactly as strongly as the supported camps do, so the wiring gate can protect a pluralism but cannot tell a supported school from a merely committed one—the data's verdict on an unsupported theory lives in the means, and in the strain and reliability reads that live on them. This refuted our own prediction, and it is the right caveat on every wiring-gate claim.

D The crisis and discovery checks, derived from the ledger

The per-round loop of §3.5 contains two threshold events: the *crisis check* (when does an agent prune the protective edges and release the hub?) and the *discovery check* (when does it wake the unconceived node?). This appendix states exactly what each check computes, derives both from the single objective the paper already carries, and lists the modelling commitments that remain after the derivation. The aim is that nothing in the two checks should read as a stipulated event: every acceptance is the model's own evidence weighed on the model's own ledger, and everything that is *not* derived is named as a commitment.

D.1 One ledger for every structural move

The bounded agent of §3.2 revises its beliefs by minimizing $D_{\mathrm{KL}}[q \,\|\, p(x \mid o, G)] - \lambda\, \mathbb{E}_q\big[U^{\top}x\big]$ (Eq. 3): fidelity to the honest posterior, traded against the value of where the commitments land, at temperature λ. A structural move—pruning an edge, waking a node—changes both terms, so the same objective scores the move:

$$\mathrm{score}(M \to \tilde{M}) \;=\; \underbrace{\Delta F}_{\text{log Bayes factor}} \;+\; \underbrace{\Delta G}_{\text{epistemic gain}} \;+\lambda\; \underbrace{\Delta U}_{\text{conviction value}}, \qquad \text{accept} \iff \text{score} > 0, \tag{10}$$

with ΔF the evidence for the edited structure, ΔG the expected information gain about any newly introduced latent (zero for a prune), and $\Delta U = \mathbb{E}_{q_{\tilde{M}}}\big[U^{\top}x\big] - \mathbb{E}_{q_M}\big[U^{\top}x\big]$ the change in the conviction-field value of the commitments. Equation (10) is not a new mechanism: it is Eq. (3) evaluated at two structures and differenced. Both threshold events of the simulation are instances of it.

D.2 The crisis check is the ledger on a prune

For a prune, every term of Eq. (10) is closed-form in the linear-Gaussian setting. The evidence half is Bayesian model reduction (Eq. 5): writing the agent's accumulated likelihood deposit as $J = \Pi - \Pi_0$, $j = h - h_0$, the Savage–Dickey identity scores the swap of the prior (Π_0, h_0) for the edge-e-reduced prior $(\Pi_0^{(e)}, h_0^{(e)})$ under the same likelihood, and—this is the part the main text does not spell out—it *already exhibits the reduced posterior*,

$$\Pi^{(e)} = \Pi_0^{(e)} + J, \qquad h^{(e)} = h_0^{(e)} + j, \tag{11}$$

so the value half costs one linear solve per candidate edge:

$$\Delta U_e \;=\; U^{\top}\big(\mu^{(e)} - \mu\big), \qquad \mu^{(e)} = \big(\Pi^{(e)}\big)^{-1} h^{(e)}, \quad \mu = \Pi^{-1} h, \tag{12}$$

with $U = \mathsf{T}u$ the agent's *current* conviction field, read along its present couplings. The crisis check is then exactly the ledger: edge e is flagged iff $\Delta F_e + \lambda\, \Delta U_e > 0$, and *crisis* is the round at which both protective belt edges are flagged, upon which the agent adopts the reduced prior composed over every edge its own ledger flags (the hub released). Pruning a conviction-protective edge moves the posterior mean away from the cherished commitments, so $\Delta U_e < 0$ and the rule reproduces the intuitive form "the evidence must overcome λ times the value the edge protects."

A natural surrogate suggests itself here: the static threshold $\Delta F_e > \lambda\, v_e$ with $v_e = |U_{\text{parent}}| + |U_{\text{child}}|$, the summed conviction magnitude at the edge's endpoints—available without a solve, and with the right *form* (it is Eq. (10) with $\Delta U_e \equiv -v_e$). But it has the wrong *content*: v_e is a static stand-in for a quantity the model can compute exactly, so we compute it. The difference is not cosmetic, and its direction is itself informative: because the reduced and full posteriors share the likelihood deposit, the mean displacement of a prune—and with it $\lambda\, \Delta U_e$—*shrinks as evidence accumulates.* A data-rich paradigm is barely protected at the threshold; almost all of its persistence must come from upstream, from the conviction gate that decides which evidence is allowed to accumulate at all (Eq. 4). The static surrogate would quietly overstate threshold protection; the exact ledger locates lock-in where the paper's results put it: gating plus isolation, not a high bar at the moment of decision.

D.3 The discovery check is the ledger on an expansion

Expansion cannot be closed-form (§3.2): a new node needs a real inversion, and—Stanford's problem of unconceived alternatives—the agent cannot plan toward a dimension it has not conceived. The discovery check therefore has exactly two parts, and only two:

1. **Arrival.** Candidate proposals arrive at a Poisson rate r. This is the one place a stipulated process enters, and it is a deliberate modelling commitment: a rate is the maximally agnostic model of conceiving what one has not conceived. The proposal's *direction* is read from the model's own prediction errors: a hidden hub b wired to the visible nodes with couplings c leaves, on marginalization, the rank-one footprint $-cc^{\top}/\Pi_{bb}$, so the leading eigenvector of the windowed error second moment is the best single-hub explanation of the residual. The window is an estimator horizon, not a gate.
2. **Acceptance.** The arrived proposal is accepted iff the *model log Bayes factor* of the bordered (hub-wired) model against the agent's current one is positive—that is, iff the residual genuinely loads on the proposed hub, equivalently iff the immediate Savage–Dickey prune of the new couplings would be rejected. The epistemic gain ΔG sits on the ledger too, but on the other side of the move: being positive for *any* new latent, it is what pays for attempting the (in general expensive) inversion, and it cannot keep a hub the data do not hold—an accept test that included it would wake on pure noise. Entertaining is paid by curiosity; keeping is paid by evidence. One consequence is owned rather than hidden: repeated looks give the Bayes factor a small false-discovery rate (rare noise crossings of $\Delta F > 0$ in the null world).

We regard this as a feature of the honest model—communities do occasionally adopt spurious structure—and its principled remedy is not a reinstated threshold but the wake-then-prune cycle named in the limitations, in which a spuriously woken node is pruned back by the same reduction that scored it.

A tempting third part is deliberately absent: a *trigger*, requiring the residual floor (the eigenvalue above) to clear some height, perhaps for a sustained number of steps, before the Bayes factor is consulted. Such a gate would duplicate the accept test's job with a free constant—the Bayes factor already rejects proposals the residual does not support—and its only principled reading, a budget on when to pay for the expensive inversion, is a statement about real scientific communities rather than about this simulation, where the inversion is a small log-determinant. The discovery check therefore has no tuned threshold: the Poisson rate carries Stanford's commitment, the eigenvector carries the direction, and the ledger decides. (The floor is still recorded as telemetry, because it is the visible signature of the residual the old paradigm had been absorbing.)

D.4 What remains a modelling commitment

The derivation above leaves a short, explicit list of choices that are *not* forced by the formalism. We state them because the paper's claim that "crisis, revolution and discovery are threshold crossings of the model's own evidence" is honest only if its boundary is drawn:

- **The belt conjunction.** "Crisis" is operationalized as *both* named mass-law edges flagged in the same round. The per-edge accept is derived; the conjunction over those two edges is our operational definition of having eliminated the phlogiston mass-law reading.
- **The ordering — a result, not a commitment.** The simulation attempts expansion only after an agent's own crisis, but the ordering is enforced by the evidence rather than by fiat: running the identical proposal + ledger pipeline for every pre-crisis agent at every step (without applying it), the ledger rejects 13,180 of 13,185 pre-crisis agent-steps; the five crossings are marginal ($\Delta F < 0.005$, two orders below the post-crisis acceptance evidence) and are exactly the repeated-look false-positive rate owned above, whose remedy is an audited wake-then-prune cycle. The intact paradigm absorbs the residual; there is nothing for the check to find until reduction exposes it.
- **The Poisson rate** r (swept), the prediction-error *window* (an estimator horizon), a short *warm-up* before reduction is permitted (an initialization guard), and the released hub's self-precision are quantitative choices of the same standing as the other dials of Table 1.